

FLOOD WATCH · Sustainability & Deployment Plan

The open core stays free forever. This plan is how it reaches the institutions that should run on it, and what funds the scale-out.

LIVE: stawberrymind.github.io/flood_river_watch

CODE + DATA: github.com/Stawberrymind/flood_river_watch

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01 Product

A five-module open pipeline running live today on free public infrastructure with zero accounts: the 2025 flood atlas, a decade hazard atlas, an impact engine (₹351–523 crore damage band on government DES yields), a daily district forecaster validated walk-forward at $8.1\times$ the base rate, and a 6-hourly live monitor issuing ਪੰਜਾਬੀ / हिन्दी / English alerts. 450+ automated tests; validated against Copernicus GFM, Sentinel-2 optical truth, ISRO NDEM sheets, and the government's own ground girdawari (Spearman $\rho = 0.72$).

02 The problem, priced

- The 2025 Punjab flood: ₹14,000 crore interim state loss estimate; ₹1,600 crore central assistance; ₹71 crore immediate relief (official figures, committed in-repo).
- Punjab has zero CWC flood-forecast stations, absent from CWC's own state-wise table and current lists. There is no incumbent public district-forecast layer to displace; Flood Watch fills an empty slot.
- Girdawari damage assessment deployed 2,167 patwaris for weeks of manual survey; the satellite layer that could target it runs for ~₹0.
- PMFBY insurers and agricultural lenders lack an independent, reproducible flood-loss extent.

03 Value proposition, and the AI component

For a district collector, four answers in one click: which tehsils flood repeatedly (a named list: Khadur Sahib 6 of 11 monsoons), what is flooding *now* (6-hourly satellite), what is most at risk over the *next three days* (a live district ranking, not a probability), and what the damage is worth (₹ band on government yields). The AI is the pipeline itself: a Random Forest SAR classifier, a self-labeled decade of training data, and a self-exciting daily forecaster whose retracted headline numbers are published alongside its real ones. Auditable discipline is the differentiator for a government buyer.

04 Sustainability model: three tiers

TIER	WHO PAYS	WHAT THEY GET	STATUS
Open core	nobody, free forever	everything public today: maps, forecasts, alerts, data, code	live
Institutional instances	state SDMAs / Revenue Depts (MoU or procurement)	hosted multi-state instances, SLAs, village-circle girdawari-verification exports, training	pilot-ready
Analytics services	PMFBY insurers, lenders, CSR / climate-resilience funds	independent loss-extent audits, recurrence underwriting layers	roadmap

Unit economics, honestly stated. Core infrastructure cost today is ₹0/month (public CI + open data). A per-state instance is priced against the avoided cost of manual survey targeting: even a 5% efficiency gain on one girdawari cycle (2,167 patwaris, weeks of work) dwarfs any realistic hosting fee. All pricing is an assumption to be tested with deployment partners; the open core is not contingent on any of it.

05 Roll-out

- Phase 0 (done): Punjab system live through monsoon 2026; the live forecaster activates 25 July, inside judging week; jurors can watch CI commit.
- Phase 1 (this monsoon): PSDMA + worst-hit District Collector outreach (drafted, dispatches with this submission); operational feedback; EOS-04 Indian-SAR cross-validation via NRSC Bhoonidhi.
- Phase 2 (two quarters): three-state expansion (the pipeline is a bounding box + district file; Assam and Bihar have NRSC frequency atlases but no live open district layer), Bhashini voice alerts, village-circle girdawari integration.
- Phase 3: national coverage via IndiaAI / NRSC partnerships; formal engagement with CWC on the station-gap finding.

06 Marketing & distribution

The alerts are the distribution: vernacular, free, quotable. Twenty print-ready district flood briefs ship in-repo as the door-opener artifact for collectors. The dam-headroom and station-gap findings are press-grade stories that carry the system's name into policy rooms, the space SANDRP already covers editorially, now with reproducible numbers.

07 Risks, with evidence

- Data feeds go dark. Happened: the BBMB dams stopped reporting centrally in July 2025. *Proven survivable by ablation*: detection skill does not depend on the reservoir feed.
- Single-sensor reliance. Sentinel-1 today; EOS-04/Bhoonidhi second SAR source is the Phase-1 mitigation.
- Adoption friction. The zero-login open core removes procurement as a precondition, so institutions can adopt before they contract.
- Founder concentration. Solo student builder, mitigated by design: 450+ tests, a method paper, a data-source registry, and a full verification log make the system transferable. Deployment support needs: government procurement navigation, PPP structuring.

08 Team

Solo student builder, Punjab, building during the 2026 monsoon. The repository is the CV.

Open core free forever · MIT + CC-BY-4.0 · ₹0/month core infrastructure · 450+ automated tests ·
github.com/Stawberrymind/flood_river_watch · stawberrymind.github.io/flood_river_watch · Companion to the Flood Watch synopsis.